

Investigations of Information Load in Aquatic Microecosystems

Wojciech Puchalski

From **20 March to 25 July 2015** (4 months), an experiment was conducted to evaluate the effects of a **coded aluminosilicate** in **30 variant complex aquatic microecosystems**, each with a volume of **5 L**. The microecosystems were established in transparent plastic containers and contained **12 g dry mass** of organic matter in the form of autumn-shed leaves: separately **beech, alder, reed**, and a mixed treatment composed of **4 g each** of leaves from the three species. The containers, including a treatment without leaves, were filled with water collected from several natural water bodies together with a mixture (ca. **4 g**) of the surface layer of bottom sediments from these sites. The water and sediment served as an inoculum containing diverse aquatic microorganisms, including **bacteria, cyanobacteria, algae, and aquatic microinvertebrates**, involved in **organic matter decomposition, primary production, and consumption**.

The microecosystems were maintained under field conditions, exposed to natural fluctuations in temperature and solar radiation, in three chemical variants: **initial acidification with hydrochloric acid to pH 4, no amendment** (natural water, initial **pH 7.4**), and an **alkalized treatment**, buffered against pH decline by the addition of **2 g calcium carbonate (CaCO₃)**.

The coded aluminosilicate was added to one half of the prepared microecosystems. Thus, each possible treatment combination was represented as a pair consisting of an **uncoded control** and a corresponding **coded variant**.

Initial water pH had no significant effect on the final amount of undecomposed leaf biomass remaining at the end of the experiment. In contrast, **leaf type strongly determined decomposition rate**, with mean mass loss ranging from approximately **22% of initial dry mass in beech** to **34-45% in the other leaf types**. The coded aluminosilicate had **no detectable effect** on the decomposition rate of beech leaves, only a **minor effect** (up to several percentage points) in reed, an effect of approximately **10 percentage points** in alder (**35-45%**), and the **strongest effect**, up to **15 percentage points**, in the mixed-leaf treatment. In the latter case, the presence of coded aluminosilicate also significantly accelerated the decomposition of beech leaves, which otherwise decomposed very slowly in the monospecific treatments. This result is consistent with earlier studies involving **Agnihotra ash**, in which the strongest effects were likewise observed in the structurally most complex microecosystems.

A characteristic feature of **alder leaf decomposition** was the release into the water of clearly visible **brown dissolved organic matter**, which formed a precipitate in the presence of calcium ions; accordingly, its concentration remained low in the alkaline treatment. In the remaining treatments, the addition of coded aluminosilicate markedly reduced the concentration of this colored dissolved material, in some cases almost eliminating it during the initial phase of the experiment, when decomposition processes predominated over primary production. This was associated with a substantially earlier onset of mass algal development in the coded alder-leaf treatments.

Overview of algal development across treatments

Leaf-free control

Acidic variant	The coded treatment supported approximately twice the algal abundance , dominated mainly by green algae.
Neutral variant	The uncoded treatment was dominated by brown cyanobacteria , whereas the coded treatment was dominated by green algae , a shift potentially more favorable for the grazing food chain.
Alkaline variant	Very low algal abundance was observed in the uncoded treatment, while the coded treatment supported numerous filamentous green algae .

Beech leaves

Acidic variant	Algal abundance was markedly higher in the coded treatment.
Neutral variant	Although overall algal abundance was similar in coded and uncoded treatments, photosynthetic activity was substantially higher in the coded variant, as reflected by elevated dissolved oxygen concentrations.
Alkaline variant	As in the neutral treatment, algal abundance remained comparable between coded and uncoded systems, but the coded treatment exhibited significantly greater photosynthetic performance , with oxygen concentrations differing by as much as 5 mg L⁻¹ (approximately 8 mg L⁻¹ in uncoded systems versus up to 13 mg L⁻¹ in coded systems).

Alder leaves

Acidic variant	The uncoded treatment developed a compact surface cyanobacterial film dominated by <i>Phormidium</i> sp., whereas the coded treatment supported green algae .
Neutral variant	The uncoded treatment formed a cyanobacterial surface scum , while the coded treatment was characterized by the presence of diatoms and abundant planktonic algae .
Alkaline variant	Algal abundance in the coded treatment was approximately 30% higher than in the uncoded

	treatment.
--	------------

Reed leaves

Acidic variant	The uncoded treatment developed a compact surface film of <i>Phormidium</i> sp., whereas the coded treatment formed a thick green algal-cyanobacterial mat , including heterocystous cyanobacteria capable of atmospheric nitrogen fixation.
Neutral variant	The uncoded treatment contained cyanobacteria and green algae in water colored by brown dissolved organic matter , whereas the coded treatment had clear water and supported cyanobacteria, green algae, and diatoms.
Alkaline variant	The uncoded treatment was characterized by a dense <i>Phormidium</i> mat, while the coded treatment supported a more diverse assemblage , including heterocystous forms.

Mixed leaf litter

Acidic variant	The uncoded treatment developed a cyanobacterial surface scum , whereas the coded treatment formed a bottom-associated green algal-cyanobacterial mat with heterocysts , accompanied by high oxygen production.
Neutral variant	The uncoded treatment was dominated by brown cyanobacteria , while the coded treatment developed a green, multispecies benthic mat .
Alkaline variant	The uncoded treatment was dominated by planktonic green algae, particularly <i>Chlamydomonas</i> sp., whereas the coded treatment supported diatoms and heterocystous cyanobacteria .

General conclusions

The addition of coded aluminosilicate frequently increased the intensity of metabolic processes, including the **decomposition of particulate and dissolved organic matter** and **algal primary production associated with oxygen release**. It often enhanced both the **taxonomic and functional diversity** of algal assemblages and potentially improved their value as food for aquatic invertebrates. In the uncoded variants, cyanobacteria often dominated; such forms are typically poorly edible or form dense, impenetrable surface films that restrict gas exchange and reduce oxygen availability within the water column and at the sediment surface. In contrast, coded variants contained substantially higher abundances of **heterocystous cyanobacteria**, which

may alleviate nitrogen limitation through **atmospheric nitrogen fixation**, thereby supporting the development of aquatic biomass.

Water reaction (**pH**) and calcium availability appeared to influence primarily the **structure of algal communities** developing in the microecosystems.

Despite the pronounced differences observed in production and decomposition rates, not all metabolic processes appeared to be stimulated by coding. Identification of the specific processes and structural shifts that can be induced with confidence by coding requires further investigation. Particularly noteworthy is the **absence of any coding effect on the decomposition of beech leaves**, which decompose slowly under natural conditions.

It is also likely that the effect of coding is relatively weak in **simple, single-component, or monospecific systems**, whereas it becomes more pronounced in more diverse systems. This may support the hypothesis that the information introduced through coding acts not primarily on individual ecophysiological processes, but rather on **interactions among species**, which are themselves based on information exchange. Such an effect could contribute to synchronization of life processes and thereby to greater metabolic efficiency at the ecosystem level.

At the level of information introduced by coding in complex systems, one may approach an explanation of the actual significance of **biodiversity**, both for the maintenance of stability and functional efficiency in natural ecosystems, which often thrive even under nutrient-poor conditions, and for **ecological agriculture** and **pond management**, where practitioners have long emphasized the positive role of biological diversity. With a broader empirical dataset, equally promising perspectives may emerge for the introduction of coded information into **forest ecosystems**, particularly during the earlier stages of silvicultural development.